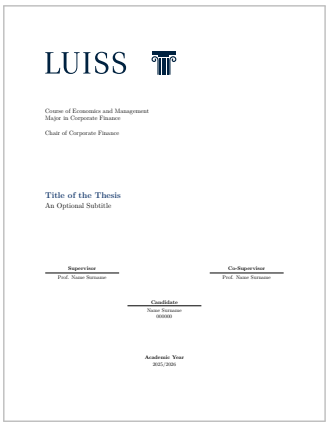
academicthesis

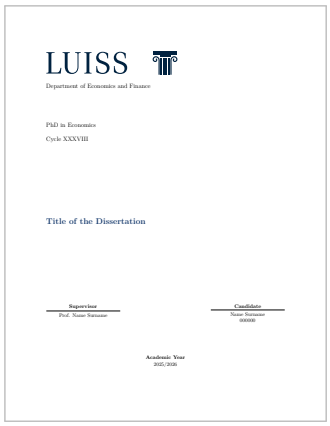
UNIVERSAL THESIS TEMPLATES FOR LATEX

LUISS Guido Carli

User and developer guide



Bachelor's and master's



PhD

Luca Salvatore

academicthesis v0.2.0 • October 2026

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Part I

User guide

For students writing a LUISS thesis: what the class does, how to set it up, every option and command available, and complete examples.

Chapter 1

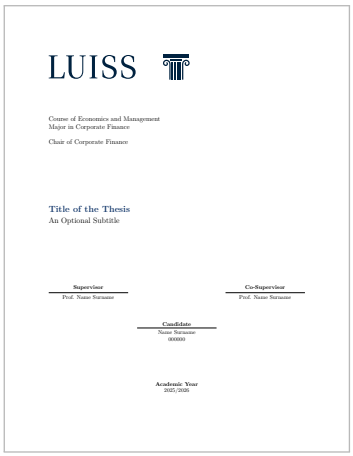
What this package does

Every LUISS thesis opens with the same cover: the university logo, the degree programme and chair, the title, the names of the supervisor and the candidate, the academic year. Reproducing it by hand means adjusting spacing, fonts and alignment, and doing it again when something changes.

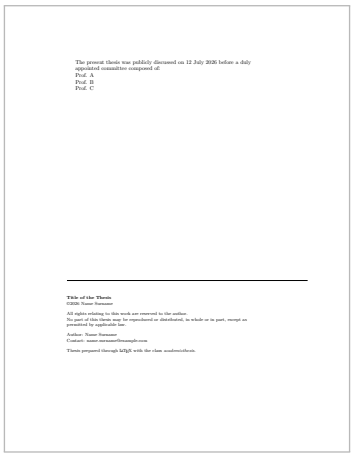
academicthesis is a LaTeX class that does this for you. You state the university and the degree, give the data of the thesis once, and the class prints the LUISS cover page, the second page with the examination details and copyright notice, and a consistent layout for the rest of the thesis:

```
\documentclass[university=luiss, master, language=english]{academicthesis}
```

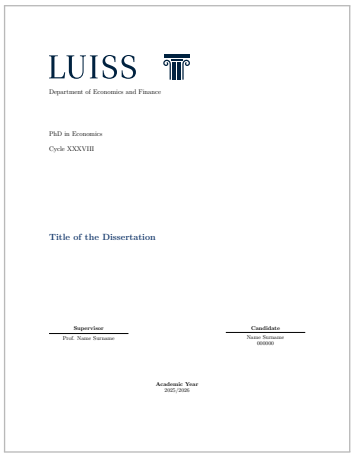
1.1 What you get



Bachelor's / master's cover



Second page



PhD cover

- **The cover page** for a bachelor's, master's or PhD thesis, with the LUISS logo and the title in LUISS blue.
- **The second page**: examination commission (or a statement that the thesis has not been defended) and a copyright notice.
- **Front-matter blocks**: abstract, statement of originality, acknowledgements and dedication, each in any of the supported languages.
- **Labels in your language**: English, Italian or French, with singular and plural forms chosen for you ("Supervisor" or "Supervisors") and, in Italian, "Candidato" or "Candidata".
- **Checks**: if a field the cover needs is missing, compilation stops with a message naming it, instead of printing a gap.
- **A bibliography** in author–year style, ready to use.

1.2 How it is organised

The class has a core shared by every university and one package per university. For a LUISS thesis the core loads the LUISS package by itself; you never load it explicitly. Part II of this guide explains the code for those who want to change it.

Chapter 2

Installing and compiling

2.1 Requirements

- A recent TeX distribution: TeX Live or MacTeX 2023 or later, or an up-to-date MiKTeX. The class is written in LaTeX3 and needs a modern LaTeX kernel.
- **pdfLaTeX** is the reference engine, used by all the examples in this guide. XeLaTeX and LuaLaTeX are also accepted (the class then loads fontspec).
- For the bibliography, biblatex and biber, both part of every full TeX distribution. Without them the class falls back to natbib and BibTeX (Section 4.2).

2.2 The files you need

academicthesis is a collection of small packages that load one another *by name*. For a LUISS thesis you need only these:

Files	Role
`core/academicthesis.cls`, `core/academicthesis-kernel.sty`	The core class, shared by every university.
`languages/academicthesis-<language>.ldf`	The words printed on the pages, one file per language (English, Italian, French).
`universities/luiss/*`	The LUISS package: its layout, labels and logos.

The other folders of `universities/` belong to other universities and can be left out.

2.3 Three ways to set it up

1. **Inside the repository.** Copy the LUISS showcase in `showcases/` and keep its first line,

   ```
   \input{academicthesis-repo-paths}
   ```

   which tells LaTeX where the package folders are. Then compile with any editor, or from a terminal with `latexmk`.
2. **Next to your thesis.** Copy the files listed above into the folder of your thesis (without sub-folders) and delete the `\input{academicthesis-repo-paths}` line. This is also how to use the class on Overleaf: upload the files to the project.
3. **Installed in your personal TeX tree**, for example `~/Library/texmf/tex/latex/academicthesis/` on macOS or `~/texmf/tex/latex/academicthesis/` on Linux. Every document on the computer can then use the class, with no extra line.

Editors

Texifier/Texpad, TeXShop, TeXstudio, VS Code (LaTeX Workshop) and Overleaf all work. In Texifier/Texpad, set the typesetter to *TeX Live (pdfLaTeX)*: its built-in engine is too old for this class.

2.4 Compiling

Run pdfLaTeX twice (for the table of contents and cross-references), with biber between the two runs when the thesis has a bibliography. `latexmk` does all of this by itself:

```
latexmk -pdf my-thesis.tex
```

Chapter 3

Writing your thesis

3.1 The shape of a thesis file

```
\documentclass[university=luiss, master, language=english]{academicthesis}

\usepackage{hyperref}            % your own packages, if any

\thesissetup{                    % the data printed on the cover
  title        = {Title of the Thesis},
  candidate    = {Name Surname},
  idnumber     = {000000},
  degree       = {Economics and Management},
  chair        = {Corporate Finance},
  supervisor   = {Prof. Name Surname},
  academicyear = {2025/2026},
  copyyear     = {2026},
}
\addbibresource{references.bib}

\begin{document}
\frontpage                       % cover page + second page

\frontmatter                     % roman page numbers
\begin{abstractthesis}[keywords = {first, second}]
  The abstract.
\end{abstractthesis}
\tableofcontents

\mainmatter                      % arabic page numbers
\chapter{Introduction}
...
\printbibliography

\backmatter
\begin{acknowledgements}
  Thanks to ...
\end{acknowledgements}
\end{document}
```

After `\frontpage`, everything is ordinary LaTeX: chapters, sections, figures, tables, equations.

3.2 Class options

All options go between the square brackets of `\documentclass`.

Option	Values	Meaning
`university`	`luiss`	Selects the LUISS package. Required.
degree	`bachelor`, `master`, `phd`	Written on its own. Chooses the cover. Default: `bachelor`.
`language`	`english`, `italian`, `french`	Language of every printed label, and hyphenation. Default: `english`. Short forms: `lang=en`, `lang=it`, `lang=fr`, or just `italian`.
`other-languages`	e.g. `{italian,french}`	Other languages used by single blocks (Section 4.1).
`no-secondpage`		Omits the second page (it is printed by default).
`discussion`		The thesis has been defended: the second page shows the date and committee, and `examdate` becomes required. Default: off.
`biblatex-options`	e.g. `{style=apa}`	Options for biblatex (Section 4.2).
anything else	`twoside`, `12pt`, `a4paper`, ...	Passed to the book class, on which a LUISS thesis is built.

LUISS has a single layout, so the `format` option is not used.

3.3 The data of your thesis

All the data is given with `\thesissetup` in the preamble, as `key = {value}` pairs. Write academic titles ("Prof.", "Dr.") yourself.

Key	Printed as	Bachelor/master	PhD
`title`	Title, in LUISS blue	**required**	**required**
`subtitle`	Under the title	optional	optional
`candidate`	"Candidate" column	**required**	**required**
`idnumber`	Under the candidate's name	optional	optional
`gender`	`male`/`female`: "Candidato" / "Candidata" in Italian	optional	optional
`degree`	"Course of ..."	**required**	—
`chair`	"Chair of ..."	**required**	—
`department`	"Department of ..." under the logo	—	**required**
`phdtitle`	"PhD in ..."	—	**required**
`cycle`	"Cycle ..."	—	**required**
`supervisor` / `supervisors`	"Supervisor(s)" column	**required**	**required**
`cosupervisor` / `cosupervisors`	"Co-Supervisor(s)" column	optional	optional
`academicyear`	"Academic Year"	**required**	**required**
`copyyear`	Second page: "© year name"	recommended	recommended
`authoremail`, `website`	Second page: contact lines	optional	optional
`examdate`	Second page: date of the defence	with `discussion`	with `discussion`
`examiners`	Second page: the committee	with `discussion`	with `discussion`

Writing the values

- `degree`, `chair`, `department`, `phdtitle` and `cycle` are written *without* the words the cover adds: `degree = {Economics and Management}` prints "Course of Economics and Management".
- `supervisor` takes one name and keeps any comma in it; `supervisors` takes several, separated by commas. The label becomes "Supervisors" by itself. The same holds for `cosupervisor(s)` and `examiner(s)`.
- `\thesissetup` can be used several times; a later value replaces an earlier one.

3.4 The major

A bachelor's or master's thesis may name a major, printed as written between the course and the chair. It has its own command; either line works:

```
\major{Major in Corporate Finance}
\luissthesissetup{major = {Major in Corporate Finance}}
```

3.5 Adjusting the cover: `\luissthesissetup`

Key	Default	Effect
`logo-file`	`logoluissblue`	Logo image, file name without extension. The plain `luiss` logo is also shipped.
`logo-width`	`0.55\textwidth`	Width of the logo.
`logo-align`	`left`	`left`, `center` or `right`.
`title-align`	`left`	Title and subtitle.
`course-align`	`left`	Course and chair (PhD: programme and cycle).
`major-align`	`left`	The major.
`department-align`	`left`	PhD: the department line.
`major`		The major (above).

All alignment keys except `logo-align` also accept `l`, `c`, `r`. For example, a centred cover:

```
\luissthesissetup{logo-align = center, title-align = center,
                  course-align = center, major-align = center}
```

3.6 The second page

`\frontpage` prints the second page after the cover, unless the class option `no-secondpage` is given.

- **Top**: without `discussion`, "The present thesis has not been defended."; with `discussion`, the defence date (`examdate`) and the committee (`examiners`).
- **Bottom**: the title, "© `copyyear` `candidate`", a reservation of rights, the author, and the e-mail and website when given.

Give `copyyear`, or the notice will have no year.

Chapter 4

Front matter and bibliography

4.1 Abstract, originality, acknowledgements, dedication

The core class provides four front-matter blocks. They are written the same way at every university; the LUISS package only sets their look.

Block	Options	Default heading (English)
`\begin{abstractthesis}`	`language`, `keywords`, `title`	Abstract
`\begin{originality}`	`language`, `title`	Statement of Originality
`\begin{acknowledgements}`	`language`, `title`	Acknowledgments
`\dedication[...]{text}`	`language`	none: the text alone on its page

- `language=` writes the block in another language: its heading, the "Keywords" label and the hyphenation follow that language. Without it, the block uses the main language of the thesis. For correct hyphenation, also list the language in the `other-languages` class option, e.g. `other-languages={italian}`; otherwise a warning reminds you.
- `keywords=` prints a list of keywords after the abstract.
- `title=` replaces the default heading.

Abstracts, statements of originality and acknowledgements are listed in the table of contents. An English abstract followed by an Italian one:

```
\begin{abstractthesis}[keywords = {corporate finance, valuation}]
  The English abstract.
\end{abstractthesis}
\begin{abstractthesis}[language = italian,
                       keywords = {finanza aziendale, valutazione}]
  L'abstract in italiano.
\end{abstractthesis}
\dedication{To my family.}
```

4.2 Bibliography

The class loads biblatex with biber for you; do not load biblatex again in your preamble. Its style is, in order of priority: the options you give with the class option `biblatex-options`; the university's default (for LUISS, author–year: `style=authoryear`); otherwise a numeric style.

```
\documentclass[..., biblatex-options={style=apa}]{academicthesis}
\addbibresource{references.bib}   % in the preamble
...
\printbibliography                % where the bibliography goes
```

If biblatex is not installed, the class warns and uses natbib with BibTEX behind the same two

commands, so the thesis still compiles.

Chapter 5

Complete examples

5.1 Bachelor's or master's thesis

```
\documentclass[university=luiss, master, language=english, twoside,
               discussion, other-languages={italian}]{academicthesis}

\major{Major in Corporate Finance}

\thesissetup{
  title        = {Title of the Thesis},
  subtitle     = {An Optional Subtitle},
  candidate    = {Name Surname},
  idnumber     = {000000},
  degree       = {Economics and Management},
  chair        = {Corporate Finance},
  supervisor   = {Prof. Name Surname},
  cosupervisor = {Prof. Name Surname},
  academicyear = {2025/2026},
  copyyear     = {2026},
  authoremail  = {name.surname@example.com},
  examdate     = {12 July 2026},              % because of `discussion'
  examiners    = {Prof. A, Prof. B, Prof. C},
}
\addbibresource{references.bib}

\begin{document}
\frontpage

\frontmatter
\begin{abstractthesis}[keywords = {first, second}]
  The abstract.
\end{abstractthesis}
\begin{abstractthesis}[language = italian, keywords = {primo, secondo}]
  Il sommario in italiano.
\end{abstractthesis}
\dedication{To my family.}
\tableofcontents

\mainmatter
\chapter{Introduction}
...
\printbibliography

\backmatter
\begin{acknowledgements}
  Thanks to ...
\end{acknowledgements}
\end{document}
```

For a bachelor's thesis write `bachelor`. Without `discussion`, leave out `examdate` and `examiners`.

5.2 PhD thesis

```
\documentclass[university=luiss, phd, language=english, twoside]{academicthesis}

\thesissetup{
  title        = {Title of the Dissertation},
  candidate    = {Name Surname},
  idnumber     = {000000},
  department   = {Economics and Finance},   % "Department of ..."
  phdtitle     = {Economics},               % "PhD in ..."
  cycle        = {XXXVIII},                % "Cycle ..."
  supervisors  = {Prof. Name Surname},
  academicyear = {2025/2026},
  copyyear     = {2026},
}

\begin{document}
\frontpage
\frontmatter
\begin{abstractthesis}
  The abstract.
\end{abstractthesis}
\tableofcontents
\mainmatter
\chapter{Introduction}
...
\end{document}
```

Complete, compilable versions are in `showcases/example-luiss-bachelor.tex` and `showcases/example-luiss-phd.tex`.

Chapter 6

Troubleshooting

Message (abridged)	Cause	Fix
Missing required thesis metadata: '*key*'	A required field is not in `\thesissetup`	Add it; the message shows an example. See Section 3.3.
Missing required thesis metadata: 'examdate'	`discussion` without the defence date	Add `examdate`, or remove `discussion`.
Missing required thesis metadata: 'chair'	Bachelor's or master's thesis without the chair	Add `chair`.
Missing required thesis metadata: 'cycle' (or 'department', 'phdtitle')	PhD data missing	Add it to `\thesissetup`.
University 'luiss' does not offer formats	A `format` option	Remove it.
File 'academicthesis.cls' not found	LaTeX cannot see the class files	Keep the line that inputs `academicthesis-repo-paths`, or copy the files next to the thesis (Chapter 2).
File 'logoluissblue' not found	Logo file missing	Copy the files of `universities/luiss/` too.
Warning: 'french' ... was not loaded by babel	A block with `language=french`	Add `french` to the `other-languages` option.
Warning: biblatex is not available	biblatex not installed	Install it, or keep the natbib fallback.

Quick reference

```
\documentclass[university=luiss, bachelor|master|phd,
               language=english|italian|french, other-languages={...},
               no-secondpage, discussion, biblatex-options={...}]{academicthesis}
\luissthesissetup{major={...}, logo-file=..., logo-width=...,
                  logo-align=left|center|right, title-align=...,
                  course-align=..., major-align=..., department-align=...}
\thesissetup{title, subtitle, candidate, idnumber, gender,
             degree, chair,                  % bachelor/master
             department, phdtitle, cycle,    % PhD
             supervisor(s), cosupervisor(s), academicyear,
             copyyear, authoremail, website, examdate, examiners}
\frontpage
\begin{abstractthesis}[language=..., keywords=..., title=...]
  ... \end{abstractthesis}
\begin{originality}[language=..., title=...] ... \end{originality}
\begin{acknowledgements}[language=..., title=...]
  ... \end{acknowledgements}
\dedication[language=...]{...}
```

Part II

Developer guide

For those who maintain or extend the code: first how the shared core works, then how the LUISS package is built on it.

Chapter 7

How the core works

This chapter describes the part of the code that every university shares. The next chapter builds on it to explain the LUISS package. The reader is assumed to know LaTeX 2ε package writing; LaTeX3 (expl3) names are explained where they matter.

7.1 Design principles

The core knows no university. `academicthesis.cls` never mentions LUISS, Politecnico di Milano or Bocconi. A university package plugs into a small number of *registries* (front pages, required fields, bibliography defaults) and *declarations* (base class, formats, second page, extra languages, extra class options) that the core consults at the right moment.

A university knows only the core. A university package uses the kernel's public commands and never another university's code, so each one can be versioned, shipped and installed on its own.

Files are found by name, never by folder. Every `\LoadClass`, `\RequirePackage`, `\input` and `\includegraphics` names a file, not a path. Once installed in a TeX tree, kpathsea finds each file whatever package shipped it; in the repository, the search path is extended instead of changing the code (Section 7.10).

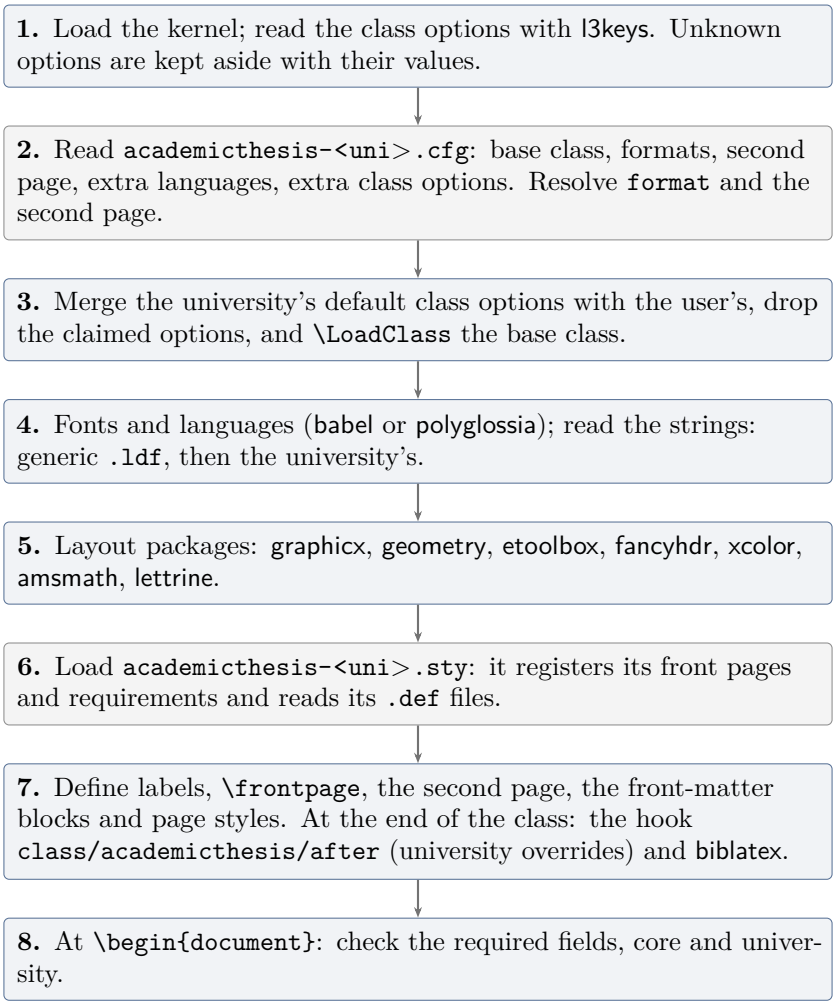
One data store, one vocabulary. All thesis data lives in a single property list with the same key names at every university. Universities change *wording* through string files, not key names.

Logic in LaTeX3, typesetting in LaTeX 2ε. Option handling, registries and validation are written in expl3; everything that puts text on the page is written in ordinary LaTeX syntax, where spaces and blank lines behave as usual (Section 7.9).

7.2 Files

File	Role
`core/academicthesis.cls`	The class: options, base class, fonts and languages, bibliography, loading the university, labels, `\frontpage`, second page, front-matter blocks, page styles, validation.
`core/academicthesis-kernel.sty`	The shared API: messages, metadata store, `\thesissetup`, lists, registries, early-configuration commands, university loader, string store. Loaded first, by the class.
`languages/academicthesis-<lang>.ldf`	Generic strings for one language (`\ProvideThesisString` calls only).
`universities/<uni>/academicthesis-<uni>.cfg`	Early configuration, read *before* the base class is loaded.
`universities/<uni>/academicthesis-<uni>.sty`	The university package proper.
`universities/<uni>/academicthesis-<uni>-*.def`	Layout code in normal LaTeX syntax, read by the `.sty`.
`universities/<uni>/academicthesis-<uni>-<lang>.ldf`	Optional: the university's wording of core strings, and its own strings.
`showcases/academicthesis-repo-paths.tex`	Lets the examples find the repository folders (not part of any package).

7.3 What happens when a thesis is compiled



Grey steps are where the university's own files run. Two consequences matter when writing a university package:

- The `.cfg` runs before `\LoadClass`: it may only use the early-configuration commands (Section 7.4) and expl3 logic, not packages or document commands.
- The `.sty` runs *before* the core defines its labels, blocks and page styles (step 7). To change one of those, a university wraps its redefinition in the hook that LaTeX runs once the class has finished:

```
\AddToHook{class/academicthesis/after}{%
  \RenewDocumentCommand\thesiskeywords{mm}{...}}
```

7.4 The early configuration file

`academicthesis-<uni>.cfg` is read with expl3 syntax on (spaces are ignored; write `~` for a space). Its commands concern only the active university, so they take no university argument.

Command	Meaning
`\DeclareThesisBaseClass {<class>}{<options>}`	Base class (book, article, ...) and its default options. A default is dropped when the user chose another option of the same group: `11pt` beats a default `12pt`, `oneside` beats `twoside`. Default: book with no options.
`\DeclareThesisFormats {<list>}{<default>}`	The values accepted by `format=`. Resolved at once, so the rest of the file can branch on `\g__academicthesis_format_tl`. A university that declares none rejects `format=`.
`\DeclareThesisSecondPage {true|false}`	Default for the second page (examination commission and copyright). The user can override it with `secondpage` / `no-secondpage`. Default: `true`.
`\DeclareThesisOtherLanguages {<list>}`	Languages always loaded with babel, besides the main one.
`\DeclareThesisClassOption {<name>}{<default>}`	Claims a `\documentclass` option for the university; it is no longer passed to the base class. `name` alone means `true`. Read it with `\academicthesis_class_option:n{<name>}` or, in normal syntax, `\thesisclassoption{<name>}`.
`\DeclareThesisCoreRequirements {true|false}`	Whether the core checks its own required fields (title, candidate, academic year, degree, supervisors). The university's requirements are checked either way. Default: `true`.

7.5 Thesis data

7.5.1 The store

All data given with `\thesissetup` goes into one global property list. `\thesissetup` is a thin l3keys front-end: each key writes one entry.

expl3 (in .sty/.cfg)	Effect
`\academicthesis_meta_set:nn {key}{value}`	Store a value.
`\academicthesis_meta_get:n {key}`	The value (expandable; empty when missing).
`\academicthesis_meta_if_empty:nTF`	Test for an empty or missing value (also `p`, `T`, `F`).
`\academicthesis_list_set:nn`, `\_set_csv:nn`, `\_add:nn`	Set a list to one item, to comma-separated items, or append one.
`\academicthesis_list_count:n`, `\academicthesis_list_if_empty:nTF`	Number of items; emptiness test.

Normal syntax (in .def)	Effect
\thesismeta {key}	The value (expandable).
\IfThesisMetaEmptyTF {key}{empty}{not empty}	Branch on a value.
\academicthesislist [sep]{list}	Typeset a list; default separator \\.
\thesislistcount {list}, \IfThesisListEmptyTF	Count items; branch on emptiness.
\IfThesisDegreeTF {bachelor|master|phd}{yes}{no}	Branch on the degree.
\thesisclassoption {name}	Value of a claimed class option.

The standard keys are title, subtitle, candidate, idnumber, degree, chair, department, major, phdtitle, cycle, academicyear, copyyear, examdate, authoremail, website, dedication, abstract, keywords, coordinator and gender, plus the lists supervisors, cosupervisors, examiners and tutors (each with a singular key that sets one item without splitting at commas). A university may store keys of its own with \academicthesis_meta_set:nn; the core never reads them.

7.5.2 Validation

At \begin{document} the core checks its own fields (unless switched off in the .cfg) and then every entry registered with

```
\DeclareThesisRequirement{<uni>}{<boolean>}{<key>}{<example>}
```

whose university is the active one and whose expl3 boolean is true: \c_true_bool (always), \g__academicthesis_bachelor_bool, _master_bool or _phd_bool. A missing field raises the error *Missing required thesis metadata: '<key>'*, whose help text shows <example>.

7.6 Registries

Command	Meaning
\DeclareThesisFrontPage {<uni>}{<degree>}{<cs>}	\frontpage looks up <uni>.<degree> and runs <cs>; if nothing is registered, it raises *No front page registered.* The command typesets its own page(s).
\DeclareThesisRequirement {<uni>}{<bool>}{<key>}{<ex>}	A required field (above).
\DeclareThesisBibDefaults {<uni>}{<options>}	biblatex options used when the user gives no biblatex-options.

7.7 Strings and languages

Every printed word comes from a string store, one per language, filled by \ProvideThesisString{<key>}{<text>}. For each language the kernel reads academicthesis-<lang>.ldf and then, if it exists, academicthesis-<uni>-<lang>.ldf.

A university file reuses the core keys to change the wording (for instance `SupervisorLabel`) and may add keys of its own.

`\thesisstring{<key>}` prints a string in the language currently selected for strings: the main language, or the one a block switched to with `language=`. It falls back to the main language, then to the key name, so a missing translation never stops compilation.

The labels `\authorlabel`, `\supervisorlabel` and `\cosupervisorlabel` choose the singular or plural string from the length of the list, and the female form of the candidate label from `gender`.

7.8 Front matter, second page, page styles

- `\frontpage` dispatches to the registered front page, then prints the second page if it is on.
- `abstractthesis`, `originality`, `acknowledgements` and `\dedication` all go through one option parser (`language`, `title`, `keywords`). Headings use `\thesisfrontchapter`, which writes `\chapter*` or `\section*` depending on whether the base class has chapters, and adds the table-of-contents entry.
- Universities can restyle without touching the parser: `\thesiskeywords{<label>}{<keywords>}` prints the keywords; `\thesisacknowledgementsbegin{<heading>}` and `\thesisacknowledgementsend` frame the acknowledgements.
- The core defines the page styles `Thesis`, `acknowledgments` and `plain`; `\frontmatter` switches to `plain` and `\mainmatter` to `Thesis`. Under a base class without chapters, the core supplies light versions of `\frontmatter`, `\mainmatter` and `\backmatter`.

7.9 Two rules for writing layout code

Typeset in normal syntax. Under `\ExplSyntaxOn` spaces are ignored and blank lines do not end paragraphs, which silently breaks any text. University packages therefore keep their expl3 code in the `.sty` and put pages in `.def` files read with the document catcodes:

```
\cs_new_protected:Npn \__myuni_input_layout:n #1
  {
    \cctab_begin:N \c_document_cctab   % no group: definitions survive
      \file_input:n { academicthesis-myuni-#1.def }
    \cctab_end:
  }
```

Never load a package from a `.def` file. When a package loaded from there ends, LaTeX restores the syntax of the class that is loading everything, expl3, and every space that follows is lost. The `.sty` loads the packages before reading the `.def`.

7.10 Search path in the repository

Inside the repository the files sit in sub-folders, which TeX does not search. Two mechanisms make them visible without any path in the code:

- **latexmk**: the root `latexmkrc` adds `core//`, `languages//` and `universities//` to `TEXINPUTS` (the trailing `//` also searches sub-folders).

- **Any editor**: `showcases/academicthesis-repo-paths.tex`, input before `\documentclass`, extends LaTeX's `\input@path` with the folders. `\input@path` does not search sub-folders, so it lists each university folder; a new university needs a line there.

7.11 Messages

All messages use the expl3 message module `academicthesis`, so they read *Class academicthesis Error* and carry a help text.

Name	When
`missing-metadata`	A required field is empty.
`missing-frontpage`	No front page registered for the university and degree.
`unknown-university`	No `academicthesis-<uni>.sty` found.
`missing-config`	The `.sty` exists but the `.cfg` does not.
`unknown-format`, `no-formats`	A `format` the university does not declare, or any `format` at a university with none.
`missing-language`	No `.ldf` for the chosen language (falls back to English).
`bibliography-fallback`	biblatex unavailable (warning).

A university defines its own messages in the same module with `\msg_new:nnnn{academicthesis}{<uni>-<name>}{...}{...}`.

7.12 Adding a university, in short

1. Create `universities/<uni>/` with `academicthesis-<uni>.cfg` (at least `\DeclareThesisBaseClass`).
2. Write `academicthesis-<uni>.sty`: `\RequirePackage {academicthesis-kernel}`, setup keys, packages, requirements, then read the `.def` files.
3. In each `.def`, define the front page and register it with `\DeclareThesisFrontPage` for each degree offered.
4. Add `academicthesis-<uni>-<lang>.ldf` files for its wording.
5. Add its folder to `showcases/academicthesis-repo-paths.tex` and a showcase.

Nothing in `core/`, `languages/` or another university changes.

Chapter 8

The LUISS package

8.1 Files

File	Role
`academicthesis-luiss.cfg`	Base class book with no default options; second page on.
`academicthesis-luiss.sty`	Setup keys, cover building blocks (expl3), loading of the cover file, requirements, bibliography defaults.
`academicthesis-luiss-bachelor-master.def`	The bachelor's and master's cover, in normal syntax.
`academicthesis-luiss-phd.def`	The PhD cover, in normal syntax.
`logoluissblue.pdf`, `luiss.pdf`	Logos.

LUISS has no `academicthesis-luiss-<lang>.ldf`: its labels are the core ones ("Supervisor", "Candidate", "Course of", "Chair of", "PhD in", "Cycle", "Department of").

8.2 The early configuration

```
\DeclareThesisBaseClass { book } { }
\DeclareThesisSecondPage { true }
```

No formats, no extra class options, the core requirements on: the user chooses paper size and sides freely.

8.3 The package, section by section

1. LUISS metadata. `major` is stored under the private key `luiss-major` through the key family `academicthesis-luiss/meta`, used by `\major`. The core never reads it. (The core's own `major` key exists for other universities; the LUISS cover reads `luiss-major`.)

2. Layout controller. `\luissthesissetup` sets the key family `academicthesis-luiss/layout`: `logo-file`, `logo-width` and the five alignment keys. Each alignment is stored in a token list (`\l__luiss_title_align_tl` and so on) and normalised to `l`, `c` or `r` by `\__luiss_norm_align:N` after every call. The old name `\LuissThesisController` is kept as an alias.

3. Building blocks. The cover is assembled from blocks written in expl3, each reading the metadata store:

Block (document-level name)	Prints
`\luiss@logobox`	The logo, at the chosen width and alignment.
`\luiss@departmentline`	"Department of `department`" (PhD).
`\luiss@programmeblock{first}{middle}{second}`	The course or programme line, an optional middle line, a gap and the chair or cycle line.
`\luiss@majorline`	The major, when given.
`\luiss@titleblock`	Title in `BlueLuiss` (RGB 21,57,108) and subtitle.
`\luiss@namesblock`	Supervisor(s), co-supervisor(s) and candidate with ID, in labelled columns; two layouts depending on whether there are co-supervisors.
`\luiss@academicyear`	"Academic Year" and the year, centred.

Fonts come from the core: `\TitleFont`, `\SubtitleFont`, `\LowerFont`. The colour `BlueLuiss` is also used for drop caps (`\LettrineFontHook`).

4. Loading the cover. The `.sty` reads exactly one `.def` with the document catcodes: `-phd.def` when the degree is `phd`, `-bachelor-master.def` otherwise.

5. Requirements.

```
% bachelor and master: degree and chair
\DeclareThesisRequirement { luiss } \g__academicthesis_bachelor_bool { degree }
    {...}
\DeclareThesisRequirement { luiss } \g__academicthesis_master_bool   { degree }
    {...}
\DeclareThesisRequirement { luiss } \g__academicthesis_bachelor_bool { chair }
    {...}
\DeclareThesisRequirement { luiss } \g__academicthesis_master_bool   { chair }
    {...}
% PhD: cycle, department, programme
\DeclareThesisRequirement { luiss } \g__academicthesis_phd_bool { cycle }      {...}
\DeclareThesisRequirement { luiss } \g__academicthesis_phd_bool { department } {...}
\DeclareThesisRequirement { luiss } \g__academicthesis_phd_bool { phdtitle }   {...}
```

The core adds title, candidate, academic year, supervisors, the degree for bachelor and master, and `examdate` with `discussion`.

6. Bibliography. Author–year by default:

```
\DeclareThesisBibDefaults { luiss } { style = authoryear, backend = biber }
```

8.4 The cover files

Both `.def` files follow the same plan and differ only in what they pass to the blocks:

```
\newcommand\luiss@frontpage@bachelormaster{%
  \begin{titlepage}%
    \thispagestyle{empty}%
    \newgeometry{left=2.75cm, right=2.75cm, top=3cm, bottom=2.5cm}%
    \luiss@logobox\par
```

```
    \vspace{0.1\textheight}%
    \luiss@programmeblock
      {\thesisstring{CourseOf} \thesismeta{degree}}%
      {\luiss@majorline}%
      {\thesisstring{ChairOf} \thesismeta{chair}}%
    \par\vfill
    \luiss@titleblock\par\vfill
    \luiss@namesblock\par
    \vspace{0.1\textheight}%
    \luiss@academicyear\par
    \vspace*{0.05\textheight}%
    \restoregeometry
  \end{titlepage}}
\DeclareThesisFrontPage{luiss}{bachelor}{\luiss@frontpage@bachelormaster}
\DeclareThesisFrontPage{luiss}{master}{\luiss@frontpage@bachelormaster}
```

The PhD page adds `\luiss@departmentline` under the logo and passes "PhD in `phdtitle`" and "Cycle `cycle`" to the programme block. The second page is the core's, not the package's.

8.5 Common changes

To change…	**Edit…**
the order or spacing of the cover	the `.def` of that degree.
how a block looks (columns, colour, rule)	the block in section 3 of the `.sty`.
a label's wording	add `academicthesis-luiss-<lang>.ldf` with `\ProvideThesisString` for the core key (e.g. `SupervisorLabel`).
which fields are required	section 5 of the `.sty`.
the default bibliography	section 6 of the `.sty`.
the default class options	the `.cfg`.

Keep the two rules of Section 7.9: text in the `.def` files, packages loaded only from the `.sty`.

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